# 1. Introduction

## 1.1 Problem Statement

Social media platforms have become a major space for discussing mental health topics and raising awareness. In Saudi Arabia, mental health influencers actively use platforms such as YouTube and TikTok to share educational content, personal experiences, podcasts, and short videos related to psychological well-being. Despite the widespread presence of such content, there is limited data-driven research that evaluates how mental health influencers impact public engagement and interaction. This creates a gap in understanding how audiences respond to mental health-related content and how influencers contribute to digital mental health awareness within the Saudi context.

## 1.2 Importance of the Project

Mental health awareness is a critical social issue that affects individuals and communities. Understanding the role and impact of mental health influencers is essential for promoting responsible communication and effective awareness strategies. This project is important because it relies on real-world, large-scale social media data collected from platforms widely used in Saudi Arabia. By analyzing influencer-generated content and audience engagement, the project provides insights that may support researchers, healthcare organizations, and policymakers in improving digital mental health initiatives.

## 1.3 Main Research Question

How does mental health influencers content in Saudi Arabia impact public engagement across social media platforms?

# 2. Literature review

## 2.1 related works

Title: **Helpful or harmful? Navigating the impact of social media influencers' health advice: insights from health expert content creators**

Authors: Kaňková, J., Binder, A., & Matthes, J.

Published in: BMC Public Health, 2024 [1]

### 1.Problem Addressed

The study addresses growing concerns about the reliability of health information shared by SMIs, who often lack professional qualifications. While misinformation (factually incorrect content) is widely studied, the authors argue that overgeneralized health messaging—content that is factually accurate but not universally applicable—can also be harmful. This is especially concerning given the close par-asocial relationships followers form with SMIs, which may lead to self-misdiagnosis or inappropriate health behaviors.

### 2.Dataset Used

Sample: 12 health experts from Czechia, including:
· 3 psychotherapists
· 3 nutritionists
· 3 medical doctors
· 2 pharmacists
· 1 physiotherapist
· Participants were aged 24–42 (M = 29.67), with Instagram follower counts ranging from 484 to 64,800.
· Recruitment was conducted via Instagram and email using profession-related keywords.

### 3. Methods/Models Applied

· Researchers conducted semi-structured interviews with 12 Czech health professionals (doctors, nutritionists, therapists) who are active on Instagram. The data was analyzed using thematic qualitative text analysis.

### 4. Key Findings

· HECCs are motivated by education, community building, and combating misinformation.
· They view non-expert SMIs negatively, though they appreciate the recent trend of including disclaimers.
· Crucially, HECCs believe overgeneralized health messaging can be just as harmful as outright misinformation, leading to self-misdiagnosis or inappropriate health choices.
· Recommended solutions include improving public literacy, educating content creators, implementing platform regulations, and verifying health professionals online.

Title: **Mental Health Professionals as Influencers on TikTok and Instagram : What role do they play in mental health literacy and help-seeking?**

Authors: Pretorius, McCashin & Coyle

Published in: Internet Interventions, 2022 [2]

## 1.Problem Addressed

Young people have low mental health literacy and low formal help-seeking despite high rates of mental disorders, and increasingly turn to social media for mental health information. This study mapped who these influencers are, what content they post, and whether it contributes to mental health literacy.

## 2. Dataset Used

50 influencer accounts (28 TikTok, 22 Instagram) owned by licensed mental health professionals with >100,000 followers, identified via platform search (psychologist, therapist, psychiatrist, psychotherapist) on 15 November 2021. Five most recent posts per account were analysed (total 250 posts: 140 TikTok, 110 Instagram).

## 3. Methods/Models Applied

Cross-sectional content analysis with dual independent screening. Deductive coding using Jorm's 7 components of mental health literacy; inductive coding for post purpose and formats. Dual coding of 10% posts with third coder verification.

## 4. Key Findings

Account characteristics: Median followers were 429k on TikTok and 493k on Instagram. Most accounts (70%) were US-based. Verified status was 21% on TikTok vs 50% on Instagram, while disclaimer/crisis info appeared on 55% of Instagram accounts but only 29% on TikTok. Eight accounts met inclusion criteria on both platforms, with 32 more present on both but qualifying on only one.
Post purpose: Education dominated both platforms (68% TikTok, 66% Instagram), though content focus differed—TikTok emphasized mental health concerns (30%), while Instagram focused on personal growth (31%). Promotion was far more common on Instagram (28% vs 3% TikTok), while engagement content was higher on TikTok (20% vs 3% Instagram).
Mental health literacy: TikTok outperformed Instagram in helping users recognize specific disorders (24% vs 7%). Instagram led slightly in promoting self-treatments (17% vs 14%) and professional help availability (9% vs 6%). Both platforms scored similarly on attitudes promoting help-seeking (TikTok 19%, Instagram 16%). Critically, neither platform had any content teaching users how to actually seek mental health information (0%).

Title: **The Impact of Social Media Influencers on Health Outcomes: Systematic Review**

Authors: Powell & Pring

Published in: Social Science & Medicine, 2024 [3]

## 1.Problem Addressed

First systematic review on whether social media influencers affect quantified health outcomes (not just attitudes). Evidence was fragmented despite influencer pervasiveness and unregulated nature, requiring synthesis to guide policy and research.

## 2. Dataset Used

· 12 peer-reviewed intervention studies (2019–2022) identified via systematic search of six databases (Medline, Embase, PsycINFO, CINAHL, Science Citation Index, Sociology Collection). Searches screened 2,489 titles, yielding 12 final studies.

## 3. Methods/Models Applied

Review methodology: PRISMA-guided systematic review with dual independent screening (>90% agreement) and narrative synthesis. Quality appraisal used the Joanna Briggs Institute Checklist for Quasi-Experimental Studies.
Eligibility criteria: Interventional designs only, with influencer as primary exposure and quantified health outcome as primary outcome. Required comparator group; peer-reviewed, English-language.
Data extraction: Dual extraction of study aims, design, population, intervention, comparator, outcomes, findings, effect sizes, and statistical methods.

## 4. Key Findings

Overall: 9/12 studies found significant influencer effects; one mixed, one null, one showed anxiety regardless of influencer status.
Dietary (n=5): Unhealthy food marketing increased children's intake; healthy food ads had no effect. Disclosures didn't help—sometimes increased intake. Kids chose healthier when unhealthy foods were promoted by sedentary/overweight influencers.
Body image (n=4): Idealized imagery increased dissatisfaction and decreased mood. Popularity didn't matter. Men affected only by bare-chested images; women affected by all.
Campaigns (n=2): Flu campaign increased vaccination (14.5% recall, P=.005). COVID-19 video alone had no effect; combined with infographics improved hand-washing only.
Anxiety (n=1): New mothers' anxiety increased after viewing idealized motherhood—same for influencers and non-influencers.
Quality: All 12 studies had bias risk; 10/12 poor generalizability; 6/12 missing effect sizes; 6/12 no power calculations; 3/12 unspecified statistical tests.

Title: **A Double-Edged Sword: Exploring Associations Between Engagement With Mental Health Content by Social Media Influencers and Psychological Well-Being**

Authors: Jaroslava Kaňková, Sofie Vranken, & Jörg Matthes.

Published in: Cyberpsychology, September 3, 2025 [4]

## 1.Problem Addressed

The study addresses a critical gap in understanding how influencer-driven mental health content affects young users. While young audiences highly trust social media Influencers, but their mental health content may have mixed effects. While it can educate and raise awareness, it may also encourage self-diagnosis, potentially harming well-being. The study explores this "double-edged sword."

## 2. Dataset Used

· Sample: 1,113 Gen Z social media users (ages 16–27) from the United States.
· Demographics: Mean age = 22.05 (SD = 2.89); 53.55% female; 46.45% male.
· Recruitment: Professional polling company (Cint) using national quotas for age, gender, and education.
· Data Collection: Cross-sectional online survey conducted July 30 – August 16, 2024.

## 3. Methods/Models Applied

Engagement with mental health content by SMIs: Self-constructed scale (liking, commenting, sharing; 3 items, ω = .81).
· Mental health literacy: Adapted from Jung et al. (2016); 4 knowledge-based items (ω = .82).
· Self-diagnostic tendencies: 3 items (ω = .82).
· Life satisfaction: 3-item version of the Satisfaction with Life Scale (ω = .80).
· Controls: Formal mental health diagnosis (yes/no), gender.
· Statistical Analysis:
· Structural equation modeling (SEM) with latent variables and mediation parameters.
· Software: R (lavaan package).
· Model fit indices: CFI = .969, TLI = .959, RMSEA = .048, SRMR = .047.

## 4. Key Findings

· Engagement was positively associated with mental health literacy.
· Literacy was positively linked to self-diagnostic tendencies.
· Self-diagnosis was negatively associated with life satisfaction.
· Engagement had a positive direct effect on life satisfaction but a negative indirect effect via self-diagnosis.
· Females reported higher self-diagnosis tendencies; those with a formal diagnosis had lower literacy and self-diagnosis scores.

## 2.2 comparison

The four studies share common themes, yet they also exhibit clear methodological differences. All examine social media influencers as health communicators and focus on related features such as engagement, mental health literacy, psychological or behavioral outcomes, credibility, and content characteristics. However, their datasets vary, ranging from cross-sectional survey data of GenZ users to manual content analysis of influencer posts, to a systematic review of intervention studies, and qualitative interviews with health experts. Thus, while the conceptual focus overlaps, the empirical foundations differ in scale, design, and level of analysis.

Their modeling approaches also differ in rigor and purpose. The 2025 study applies structural equation modeling (SEM) to test mediation mechanisms, offering an explanatory statistical framework. In contrast, the 2022 study relies on descriptive content coding, the 2024 systematic review uses PRISMA-guided narrative synthesis without meta-analysis, and the 2024 qualitative study employs thematic analysis. Collectively, these studies demonstrate that influencer health content can have both beneficial and harmful effects, and they highlight concerns about misinformation, overgeneralization, and credibility. However, they share limitations: reliance on cross-sectional or short-term designs, heavy dependence on self-report or manual coding, predominantly Western samples, and limited generalizability. Importantly, none integrates influencer attributes, content features, engagement behavior, and mental health outcomes within a unified computational framework, nor do they apply large-scale predictive or machine learning approaches.

Our data science project builds upon existing research on social media engagement and mental health by offering a structured, quantitative analysis within the Saudi context. Rather than focusing broadly on general engagement patterns, it examines how specific content characteristics—such as themes (educational, personal experiences, podcasts), video length, keywords (e.g., “podcasts” vs. “tips”), and influencer tier—drive measurable interaction outcomes like likes, views, and comments. It also introduces a cross-platform comparison between YouTube and TikTok to identify differences in audience behavior and content performance. Through this focused, data-driven approach, the project extends existing literature by systematically evaluating how content features shape engagement patterns in Saudi mental health social media.

# 3. Data Sources

## 3.1 Data Source #1: Youtube Public Data

The first data source consists of publicly available data collected from YouTube using the official YouTube Data API. YouTube was selected because it hosts a wide range of long-form mental health–related content, including interviews, podcasts, awareness discussions, and educational videos shared by mental health influencers and professionals in Saudi Arabia.

The collected data includes unstructured textual data, such as video titles and descriptions, in addition to structured engagement-related metadata.

- Platform: YouTube
- Collection Method: API-based data collection
- Data Type: Unstructured textual data with metadata

## 3.2 Data Source #2: TikTok Public Data

The second data source consists of publicly available TikTok data collected using an API-based service. TikTok was selected due to its popularity among younger audiences in Saudi Arabia and its role in spreading short-form mental health awareness content through influencers.

The collected data includes unstructured textual data, such as video captions and descriptions, along with engagement metrics that reflect audience interaction.

- Platform: TikTok
- Collection Method: API-based data collection
- Data Type: Unstructured textual data with metadata

To provide a clearer understanding of the collected data, this Table summarizes the main characteristics of the two primary data sources used in this study.

| Attribute | Youtube Dataset | TikTok Dataset |
|---|---|---|
| Data Type | Public video-level data | Public video-level data |
| Content Format | Long-form videos | Short-form videos |
| Main Text Fields | Title, Description | Caption, Description |
| Engagement Metrics | Views, Likes, Comments | Views, Likes, Comments, Shares |
| Language | Arabic and English | Arabic and English |
| Geographic Focus | Saudi Arabia | Saudi Arabia |
| Collection Method | Youtube Data API | API-based service |

Both datasets contain a combination of structured and unstructured data. The structured data includes numerical engagement metrics that reflect audience interaction, while the unstructured data consists of textual content such as video titles, descriptions, and captions.

These two platforms were selected because they represent complementary forms of content. YouTube provides longer and more detailed discussions, interviews, and educational videos, while TikTok offers short-form content that is highly popular among younger audiences in Saudi Arabia.

Together, these datasets provide a comprehensive view of how mental health influencers create content and how audiences engage with that content across different social media platforms.

## 3.3 Dataset Details

The final dataset was constructed by collecting data from YouTube and TikTok and merging both sources into a unified dataset for analysis.

- Platforms Used: YouTube and TikTok
- Data Type: Unstructured textual data
- Language of Data: Arabic
- Collection Methods: API-based data collection
- Data Formats: CSV, Excel, JSON, and HTML

**Observations (Rows)**

- YouTube dataset: 3,336 rows
- TikTok dataset: 403 rows
- Combined dataset: 3,739 rows in total

**Main Features**

| ***Column Name*** | ***Description*** |
|---|---|
| **video_id** | Unique identifier for each video |
| **video_url** | Link to the video |
| **title** | Title of the video |
| **description** | Description of the video |
| **channel_id** | Unique identifier for the channel |
| **channel_title** | Name of the channel |
| **subscriber_count** | Number of subscribers of the channel |
| **influencer_tier** | Category of influencer (Nano, Micro, Macro) |
| **search_query** | Keyword used to collect the video |
| **published_at** | Date the video was published |
| **view_count** | Number of views |
| **like_count** | Number of likes |
| **comment_count** | Number of comments |
| **duration** | Duration of the video |

| | |
|---|---|
| **total_views** | Total views of the channel |
| **video_count** | Total number of videos in the channel |
| **source** | Platform source |
| **data_type** | Type of data |

## 3.4 **Search Query Design**

To ensure comprehensive coverage of mental health–related social media content in Saudi Arabia, a structured set of search queries was designed for both YouTube and TikTok. The queries were developed in both Arabic and English to capture culturally relevant local discussions as well as broader regional and international content related to the Saudi context.

### *3.4.1 YouTube Search Queries*

A total of 105 YouTube search queries were developed and organized into thematic categories to comprehensively capture mental health-related content within the Saudi context, as well as broader English-language content relevant to analysis. The queries were designed to reflect influencers, social impact, clinical perspectives, cultural factors, national initiatives, and demographic variations in mental health discourse.

*Category 1 - Saudi Influencers (9 queries):* مؤثري الصحة النفسية السعودية، بودكاست نفسي سعودي، صناع محتوى الصحة النفسية سعوديين، مشاهير التوعية النفسية السعودية، سفراء الصحة النفسية السعودية، أخصائي نفسي سعودي يوتيوب، تجارب الصحة النفسية في السعودية، مبادرات الصحة النفسية المملكة، تفاعل السعوديين مع الصحة النفسية

*Category 2 - Social Impact (8 queries):* تأثير المشاهير على الصحة النفسية السعودية، حملات التوعية النفسية السعودية، مؤثرين سعوديين يتحدثون عن الاكتئاب، نصائح مؤثرين عن القلق السعودية، تأثير السوشيال ميديا على نفسية السعوديين، قصص تعافي نفسية سعودية، توعية نفسية مشاهير السعودية، لقاءات معالجين نفسيين سعوديين

*Category 3 - English (10 queries):* Mental Health Influencers Saudi Arabia, Saudi Mental Health Content Creators, Mental Health Awareness Campaigns KSA, Saudi Psychology Podcasts, Mental Health Social Media Impact Saudi, Psychiatrist Interviews Saudi Arabia, Mental Health Activists Saudi Arabia, Digital Mental Health Saudi Arabia, Saudi Youth and Mental Health Awareness, Top Mental Health Channels Saudi Arabia

*Category 4 - Specific Mental Health Conditions (12 queries):* الاكتئاب في السعودية، القلق النفسي سعودي، الوسواس القهري السعودية، اضطراب ثنائي القطب سعودي، نوبات الهلع سعودية، الاحتراق الوظيفي السعودية، الصدمة النفسية سعودية، الوحدة والعزلة سعودي، الاكتئاب بعد الولادة سعودية، فوبيا اجتماعية سعودي، اضطراب النوم نفسي سعودي، إدارة الغضب السعودية

*Category 5 - Demographic Groups (8 queries):* صحة نفسية شباب سعودي، صحة نفسية المرأة السعودية، نفسية الطفل السعودي، صحة نفسية المراهقين السعودية، ضغوط الأم السعودية نفسياً، طالب جامعي نفسية السعودية، الموظف السعودي ضغط نفسي، صحة نفسية كبار السن السعودية

*Category 6 - Cultural & Social Context (8 queries):* وصمة الصحة النفسية السعودية، ضغط المجتمع نفسية سعودية، الصحة النفسية والدين السعودية، العلاقات والصحة النفسية سعودية، قصص تعافي نفسية سعودية يوتيوب، الطلاق وتأثيره النفسي سعودي، التنمر في السعودية نفسياً، الغربة والحنين سعودي

*Category 7 - National Initiatives (6 queries):* مبادرة الصحة النفسية رؤية 2030، البرنامج الوطني للصحة النفسية، خط مساندة النفسي السعودي، جمعية الطب النفسي السعودية، حملات وزارة الصحة النفسية السعودية، تطبيق صحة نفسية سعودي

*Category 8 - Content Types (9 queries):* نصائح نفسية يومية سعودية، تمارين نفسية سعودية، تأمل ومايندفولنس سعودي، علاج سلوكي معرفي سعودي، تطوير الذات النفسي السعودية، الذكاء العاطفي سعودي، استشارة نفسية سعودية، جلسة علاج نفسي سعودية، محتوى نفسي هادف سعودي

*Category 9 - English - Specific Mental Health Conditions (10 queries):* Depression Awareness Saudi Arabia, Anxiety Disorder Saudi Arabia, OCD Treatment Saudi Arabia, PTSD Saudi Arabia, Burnout Saudi Employees, Postpartum Depression Saudi Women, Social Anxiety Saudi Arabia, Panic Attacks Saudi Arabia, Bipolar Disorder Saudi Arabia, Stress Management Saudi Arabia

*Category 10 -English - Demographic Groups (7 queries):* Saudi Women Mental Health, Saudi Men Mental Health Stigma, Saudi Teenagers Mental Health, Saudi University Students Mental Health, Saudi Youth Mental Health Awareness, Saudi Mothers Mental Health, Saudi Expat Mental Health

*Category 11- English - Cultural & Social Context (8 queries):* Mental Health Stigma Saudi Arabia, Seeking Therapy Saudi Arabia, Mental Health Taboo Saudi Arabia, Islam and Mental Health Saudi Arabia, Saudi Culture Mental Health, Social Media Impact Mental Health Saudi Arabia, Mental Health Misinformation Saudi Arabia, Mental Health Awareness Saudi Society

*Category 12 -English - National Initiatives & Campaigns (8 queries):* Mental Health Vision 2030 Saudi Arabia, Saudi National Mental Health Program, KSA Mental Health Campaign, Saudi Ministry of Health Mental Health, Mental Health Policy Saudi Arabia, Mental Health App Saudi Arabia, Saudi Mental Health Hotline, Mental Health NGO Saudi Arabia

*Category 13 -English - Impact Assessment (10 queries):* Impact of Mental Health Influencers Saudi Arabia, Mental Health Influencer Credibility Saudi, Mental Health Content Quality Saudi Arabia, Misinformation Mental Health Social Media Saudi, Mental Health Help Seeking Saudi Arabia, Online Mental Health Resources Saudi Arabia, Mental Health Awareness Effectiveness Saudi, Influencer Mental Health

Advice Saudi Arabia, Public Perception Mental Health Saudi Arabia, Mental Health Discourse Saudi Social Media

**Rationale**

Arabic queries were prioritized because they reflect authentic local discussions and how Saudi users naturally search for mental health content. English queries were included to capture bilingual and internationally oriented content discussing the Saudi mental health context. The combination of both languages enabled broader coverage of awareness campaigns, professional advice, personal recovery stories, podcasts, and discussions related to mental health conditions such as anxiety and depression.

### *3.4.2 TikTok Search Queries*

A total of 11 search queries were designed for TikTok.

- الصحة النفسية السعودية
- أخصائي نفسي سعودي
- مستشار نفسي سعودي
- توعية نفسية تيك توك
- نصائح نفسية سعودية
- عيادات نفسية الرياض
- استشارات نفسية جدة
- طبيب نفسي المملكة
- تطوير الذات سعودي
- صحتك النفسية السعودية
- أخصائية نفسية الرياض

**Rationale**

Fewer queries were used for TikTok because TikTok's recommendation algorithm tends to provide more diverse results per query compared to YouTube. In addition, city-specific queries such as Riyadh and Jeddah were included to capture geographically relevant content from major Saudi cities where mental health professionals and awareness initiatives are more active.

## 3.5 Data Collection and Processing

Data collection was conducted using API-based methods to ensure structured, ethical, and reproducible access to publicly available social media data.

For YouTube, the YouTube Data API v3 was implemented in Python using the google-api-python-client library within a Google Colab environment. Pagination and batch processing techniques were used to efficiently retrieve large volumes of video metadata and engagement statistics.

For TikTok, data collection was performed through an API-based service accessed via RapidAPI using Python's requests library. The collected data included captions, creator information, engagement metrics, and publication dates.

Several preprocessing and cleaning steps were applied before constructing the final dataset:

- Duplicate removal using both set-based tracking and pandas.drop_duplicates()
- Standardization of Arabic text encoding using utf-8-sig
- Handling missing engagement metrics using default values and conditional validation
- Mapping platform-specific schemas into a unified structure
- Adding source-tracking columns to preserve data lineage
- Exporting the processed dataset into CSV, Excel, JSON, and HTML formats

These steps ensured consistency, usability, and compatibility of the final dataset for later analysis phases.

## 3.6 Data Collection Summary

The final dataset consisted of publicly available mental health–related social media content collected from both YouTube and TikTok.

- YouTube dataset: 3,336 rows
- TikTok dataset: 403 rows
- Combined dataset: 3,739 rows

Each row in the dataset represents a unique social media content item related to mental health awareness, influencers, psychological counseling, or public engagement within the Saudi context.

## 3.7 Bias Evaluation

**Representation Bias:**

The dataset represents users who actively engage with mental health content on YouTube and TikTok. This may overrepresent younger and more digitally active users in Saudi Arabia while underrepresenting individuals who are less active on social media platforms.

**Measurement Bias:**

Engagement metrics such as views, likes, comments, and shares may not fully reflect genuine understanding or positive mental health impact. Users may interact with content due to entertainment value, trends, or algorithmic exposure rather than meaningful awareness.

**Historical Bias:**

Some collected content may reflect societal attitudes toward mental health from earlier periods, which may differ from current perspectives. This historical context could influence the interpretation of engagement patterns.

# 4. Objectives

The primary goal of this research is to evaluate the digital mental health landscape in Saudi Arabia by analyzing influencer-generated content and its resonance with the public. Specifically, the project aims to:

- Analyze Content-Driven Engagement: Identify how different content themes (e.g., educational, personal experiences, or podcasts) influence public interaction metrics (Likes, Views, and Comments).
- Engagement Dynamics: Identify the key factors (such as video length, specific search keywords like "podcasts" vs "tips", or influencer tier) that act as catalysts for public interaction (Likes, Comments, and Shares).
- Cross-Platform Comparative Analysis: Determine how audience behavior and content themes differ between YouTube (long-form/educational) and TikTok (short-form/engaging) within the Saudi context.
- Thematic Landscape Identification: Discover the most prevalent mental health topics discussed (e.g., Depression, Anxiety, or Self-development) and evaluate which topics generate the highest "depth of engagement" (comment-to-view ratio).

# 5. Method

To achieve the research objectives, a structured data science pipeline will be implemented. The methodology follows a quantitative and descriptive analytical approach:

## 5.1 Data Integration & Preprocessing

- Merging Datasets: Consolidating YouTube (3,336 records) and TikTok (377 records) into a unified dataframe while maintaining a "Platform" tag for comparative analysis.
- Text Cleaning: Using Python libraries (such as Pandas and Re) to clean titles and descriptions by removing stop-words, emojis, and special characters, while preserving relevant Saudi dialectal nuances and English medical terms (Code-switching).
- Handling Missing Values: Auditing engagement metrics to ensure data integrity, particularly for videos with disabled comments or hidden like counts.

## 5.2 Feature Engineering & Categorization

- Engagement Normalization: To ensure fair comparison between platforms, we will calculate the Engagement Rate (ER) and Depth of Engagement (DoE):
- Thematic Labeling (Rule-Based): Categorizing content into themes (e.g., Depression, Anxiety, Podcasting, Self-help) using a keyword-matching algorithm based on the 57 predefined search queries.
- Influencer Stratification: Segmenting creators into tiers (Nano, Micro, Macro) based on their follower/subscriber counts to analyze the correlation between reach and interaction quality.

## 5.3 Analytical Framework

- Descriptive Statistics: Summarizing the distribution of mental health topics across Saudi social media to identify the most discussed vs. most engaging themes.
- Correlation & Statistical Testing: Applying Spearman’s Rank Correlation to determine the relationship between influencer tiers, video metadata (like titles), and engagement outcomes.
- Comparative Cross-Platform Analysis: Using visualization techniques (e.g., Boxplots and Bar charts) to contrast how the Saudi audience interacts with mental health content on YouTube versus TikTok.

# 6. Challenges and Recommendations

The data collection and integration phase involved several technical and structural obstacles that required strategic mitigation. A primary challenge was navigating the

strict API rate limits and daily quota constraints, particularly with the YouTube Data API. This necessitated the implementation of efficient batch processing and optimized query designs to ensure comprehensive data retrieval within the allowed limits. Additionally, handling Arabic text encoding presented significant hurdles; maintaining robust UTF-8 support was essential to prevent data corruption when storing bilingual content and various Saudi dialectal nuances. The inherent heterogeneity of the data schemas between the YouTube and TikTok APIs also required complex mapping to create a unified and consistent dataset. Furthermore, intermittent network connectivity during extended collection sessions occasionally interrupted API calls, an issue that was addressed by developing custom retry logic and robust error-handling mechanisms to ensure data continuity.

To mitigate these challenges and enhance future data collection efforts, several strategic recommendations are proposed. Future research should prioritize the implementation of asynchronous processing and automated data streaming to optimize retrieval speeds and handle quota limits more efficiently. From an analytical perspective, integrating advanced Arabic Natural Language Processing (NLP) frameworks, specifically specialized lemmatizers for Saudi dialects, would further refine the text pre-processing stage and reduce linguistic noise. Additionally, developing cross-platform influencer mapping techniques would allow for a more holistic evaluation of an influencer's impact across different social media ecosystems. Finally, expanding the scope to include multi-modal analysis, such as video-to-text transcription and visual sentiment recognition, would provide a deeper understanding of how non-verbal cues contribute to digital mental health advocacy in the Saudi context.

# 6. Data Cleaning and Preprocessing

## 6.1 Overview

Data cleaning and preprocessing are among the most critical stages in any data science project, as they ensure the quality and reliability of the data used for modeling and exploratory analysis. In this project, two independent datasets were processed: primary data collected from YouTube and TikTok, and secondary data related to mental health indicators among social media users. The preprocessing stage addressed several challenges, including cross-platform integration, Arabic text cleaning, and feature engineering.

## 6.2 Primary Data Processing (YouTube and TikTok)

### 6.2.1 Loading the Data and Separating Platforms

The raw data were imported from the merged file master_saudi_health_impact.csv using the Pandas library. Since the dataset contained records from both platforms in a single file, the first step involved separating the records according to the platform

column. TikTok records were identified by the value 'TikTok', while YouTube records were identified by missing values (NaN) in the same column because the platform label had not been assigned during data collection. This separation resulted in 377 TikTok records and 3,336 YouTube records.

### 6.2.2 Column Selection and Schema Standardization

Due to differences in the data schemas between the two platforms, a subset of relevant columns was selected separately for each platform. For TikTok, the selected columns included: author_nickname, description, likes, comments, shares, views, date, and search_query. For YouTube, the selected columns included: channel_title, title, description, view_count, like_count, comment_count, subscriber_count, influencer_tier, published_at, duration, total_views, video_count, engagement_rate, and search_query. A constant platform column was then added to each dataset to preserve platform identity for later comparative analysis.

### 6.2.3 Data Type Correction

For the YouTube dataset, the published_at column was converted into datetime format using pd.to_datetime() with errors='coerce' to handle invalid values. All engagement metrics were converted into numeric data types. The duration column required special processing because it stored values in ISO 8601 format (e.g., PT6M47S). A custom function was therefore developed to extract hours, minutes, and seconds and convert them into decimal minutes, after which the original column was removed. For TikTok, the date column and the four engagement-related columns were converted into their appropriate data types.

### 6.2.4 Handling Missing Values

Different strategies were applied depending on the nature of each column. Missing descriptions were replaced with the default text 'No Description'. Missing engagement metrics (likes, comments, and shares) were filled with zeros, while records lacking view counts were removed entirely. The rationale behind replacing engagement metrics with zeros is that zero values often indicate disabled comments or hidden likes rather than truly missing information. In contrast, records without view counts could not be used in engagement rate calculations.

### 6.2.5 Arabic Text Cleaning

A clean_arabic_text() function was applied to description and title fields using a structured sequence of preprocessing steps:

1. Removing hashtags to reduce noise in textual analysis.

2. Removing emojis across multiple Unicode ranges.
3. Removing user mentions (@username).
4. Removing URLs.
5. Removing bidirectional control characters commonly found in Arabic text that may cause processing errors.
6. Normalizing repeated punctuation marks.
7. Compressing extra whitespace.

In addition, a custom function was applied to remove phone numbers in different formats.

### 6.2.6 Feature Engineering

The engagement rate was calculated by dividing the total interactions (likes + comments, plus shares for TikTok) by the number of views and multiplying the result by 100. YouTube video durations were categorized into four groups:

- **Short** (≤ 1 minute)
- **Medium** (1–5 minutes)
- **Long** (5–20 minutes)
- **Very Long** (> 20 minutes)

Because the TikTok API does not directly provide video duration, a custom estimation function was developed based on description length, content category, influencer tier, and engagement rate, with logical bounds between 0.5 and 10 minutes. Year and month variables were also extracted from date columns for temporal analysis.

Regarding influencer classification, the influencer_tier variable was already available through the YouTube API. For TikTok, cumulative views per creator were calculated and categorized into:

- **Mega Influencer** (≥ 10 million views)
- **Macro Influencer** (1 million–9.9 million views)
- **Micro Influencer** (100 thousand–999 thousand views)
- **Nano Influencer** (10 thousand–99 thousand views)
- **Regular Creator** (< 10 thousand views)

6.2.7 Merging the Platforms into a Unified Dataset

After preprocessing each platform independently, the columns were renamed into a unified schema such as creator_name, views, likes, and published_date. The two datasets were then merged into a single combined dataframe containing 3,739 records. Columns unavailable in one platform were filled with NaN values to maintain schema consistency. For example, the shares column is unavailable in YouTube because the

API does not expose it, while subscriber_count, duration_minutes, and duration_category are naturally absent in TikTok.

### 6.2.8 Topic Classification

A custom assign_topic() function was developed to map search queries into five thematic categories using predefined keyword lists:

- **Anxiety Disorders** (anxiety, panic, phobia, obsessive-compulsive)
- **Mood Disorders** (depression, bipolar, trauma)
- **Mental Health and Society** (loneliness, burnout, anger, sleep)
- **Specialized Mental Health Creators** (psychiatrist, podcast, specialist)
- **Awareness and Public Mental Health** (mental health, wellbeing, self-development)

Queries that did not match any category were labeled as "Other" for later review.

### 6.2.9 Sentiment Analysis

A rule-based sentiment analysis approach was implemented using domain-specific keyword dictionaries. Positive sentiment included words such as *recovery*, *hope*, *success*, *treatment*, and *support*. Negative sentiment included words such as *depression*, *anxiety*, *loneliness*, *trauma*, and *burnout*. Neutral sentiment was assigned as the default category when neither positive nor negative indicators dominated.

### 6.2.10 Final Missing Value Handling and Duplicate Verification

Before saving the final dataset, a final preprocessing pass was conducted. Missing creator names were replaced with 'Unknown', and missing influencer tiers were also replaced with 'Unknown'. Additionally, NaN values in the shares column for YouTube records were restored instead of being replaced with zero in order to explicitly distinguish between unavailable data and actual zero shares. Duplicate detection was performed using a composite identifier consisting of five columns: creator_name, description, platform, views, and likes, allowing true duplicates to be distinguished from merely similar videos.

### 6.2.11 Language Detection

A language column was added by classifying each textual description according to the proportion of Arabic and English characters. A record was labeled as **Arabic** if Arabic characters constituted at least 80% of the text, **English** if English characters constituted at least 80%, and **Mixed** otherwise. The final dataset was saved in CSV format using utf-8-sig encoding to ensure proper Arabic language support.

# 7. Exploratory Data Analysis (EDA)

## 7.1 Primary Data

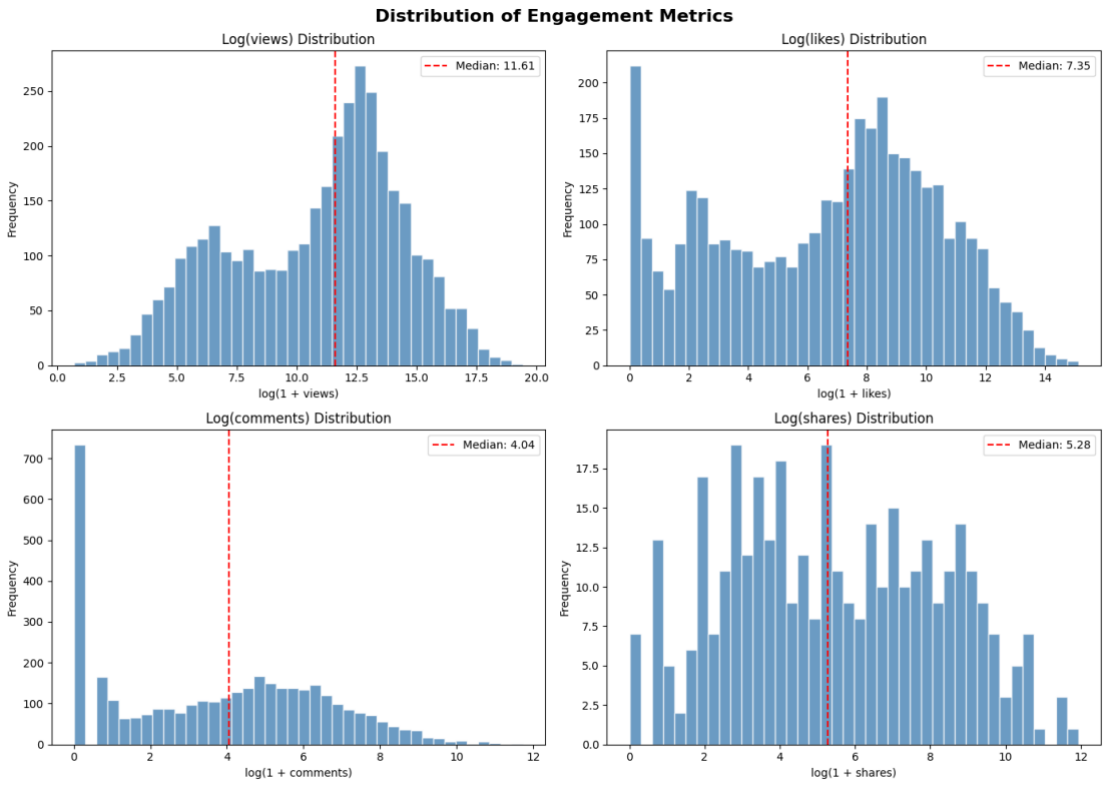
### 7.1.1 Overview of the Dataset

The combined dataset consists of **3,713 records** collected from two major social media platforms: YouTube (3,336 records) and TikTok (403 records). The dataset includes both structured engagement metrics — such as views, likes, comments, and shares — and unstructured textual data in the form of video titles, descriptions, and captions written primarily in Arabic. Initial inspection revealed that the data types were consistent across columns, with engagement metrics stored as numeric values and textual fields stored as string objects. A platform tag was preserved throughout all analyses to enable cross-platform comparisons.

### 7.1.2 Missing Values and Data Quality

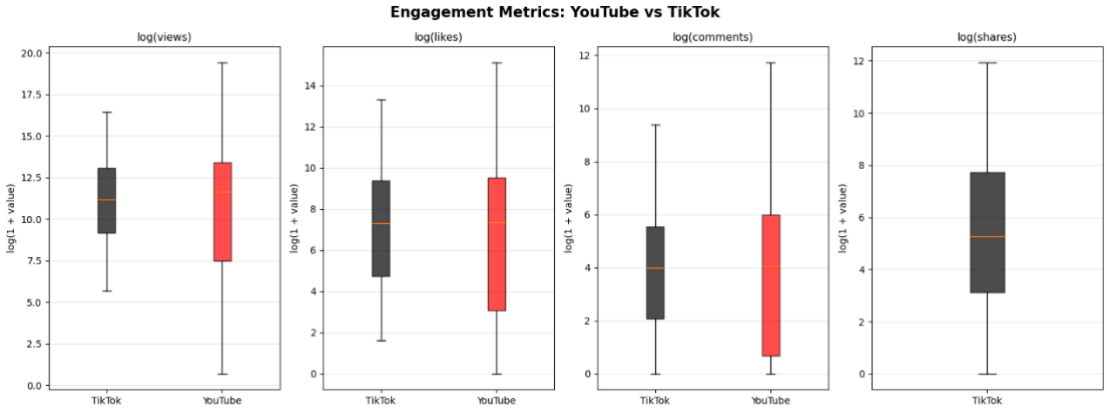
Before conducting any analysis, the dataset was audited for missing values and inconsistencies. The most notable missing data occurred in the **shares** column for YouTube records, which is expected given that the YouTube Data API does not expose public share counts. Additionally, a subset of videos had zero values in the comments column, likely reflecting videos with disabled comment sections rather than genuine absence of engagement. These cases were flagged and handled appropriately during feature engineering. Overall, the dataset demonstrated acceptable data quality, with missing values concentrated in non-critical fields.

## 7.1.3 Univariate Analysis




To understand the distribution of each engagement metric, histograms were plotted for views, likes, comments, and shares. Due to the heavily right-skewed nature of social media data — where a small number of viral videos attract disproportionately large engagement — a **log(1 + x) transformation** was applied to all metrics before visualization. The transformed distributions revealed that most videos cluster around low-to-moderate engagement levels, with a long tail of high-performing outliers. This pattern is consistent with the well-documented power-law distribution of social media content performance.

### 7.1.4 Platform Comparison



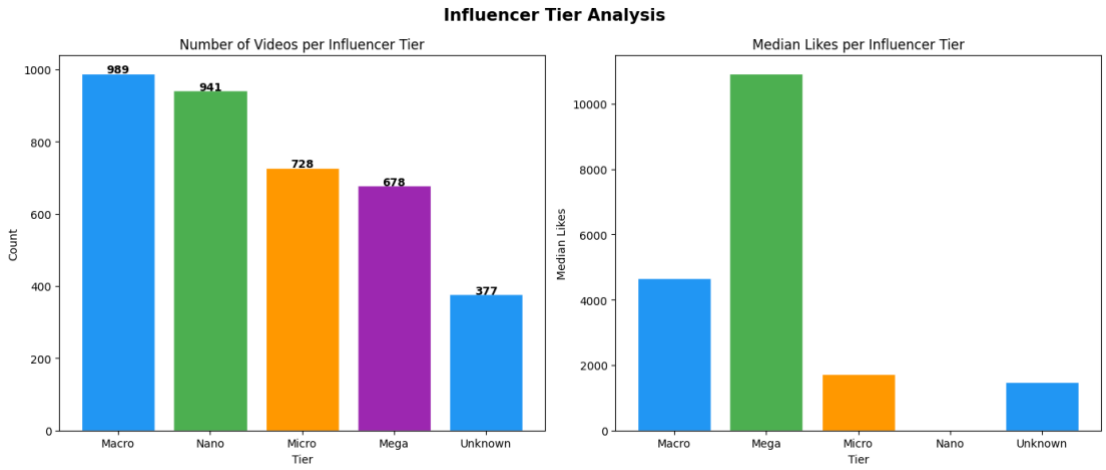

A key objective of this analysis was to compare engagement patterns between YouTube and TikTok. Boxplots were generated for each engagement metric, grouped by platform, using log-transformed values to account for scale differences. To determine whether the observed differences were statistically significant, the **Mann-Whitney U test** was applied — a non-parametric test appropriate for non-normally distributed data.

The results indicated that TikTok content, despite representing only 11% of the total dataset, achieved comparably high engagement rates relative to YouTube. This suggests that TikTok's short-form format may amplify audience interaction efficiency. Statistically significant differences ($p < 0.05$) were observed in the likes and views distributions between the two platforms.

### 7.1.5 Influencer Tier Analysis

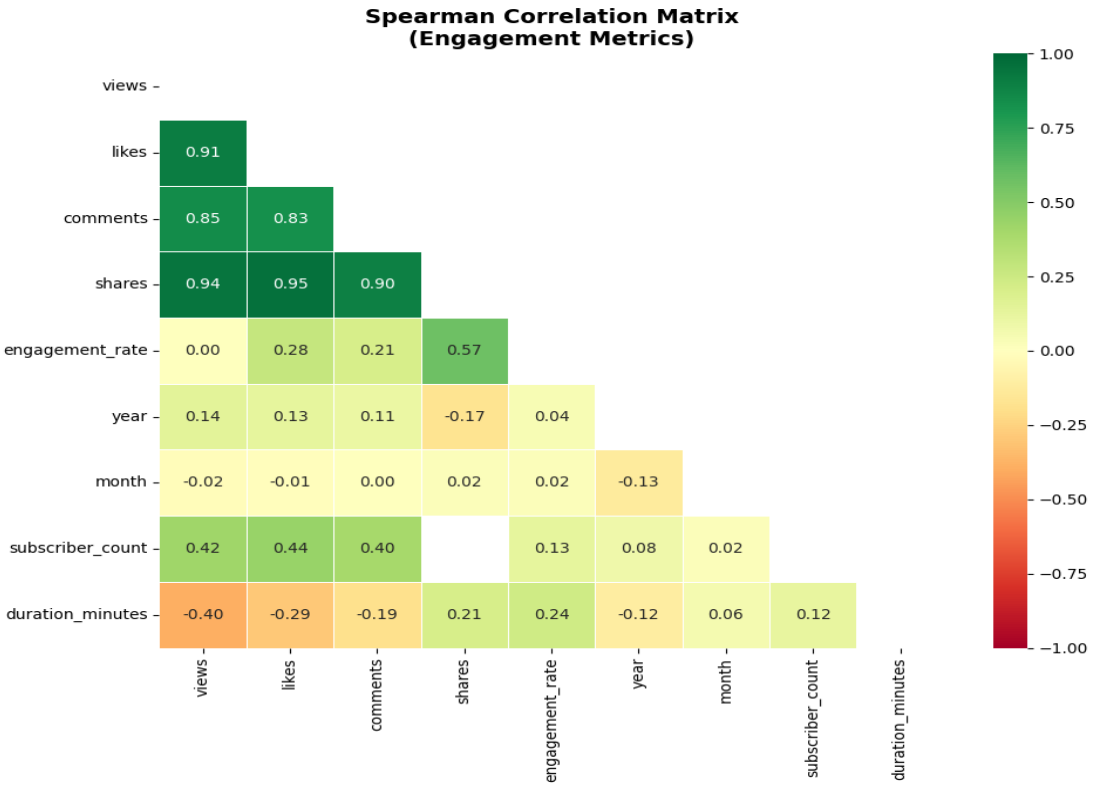
Influencers were stratified into four tiers based on their follower or subscriber counts:

| *Tier* | *Follower Range* |
|---|---|
| *Nano* | < 10,000 |
| *Micro* | 10,000 – 99,999 |
| *Macro* | 100,000 – 999,999 |
| *Mega* | ≥ 1,000,000 |




The distribution of content across tiers was visualized alongside median likes per tier. The analysis revealed that **Macro influencers** contributed the largest volume of reinforcing the ,content in the dataset and also generated the highest median likes Micro influencers ,However .relationship between reach and engagement volume suggesting ,demonstrated competitive engagement rates relative to their audience size more niche creators may foster stronger audience connections in the ,that smaller mental health space.

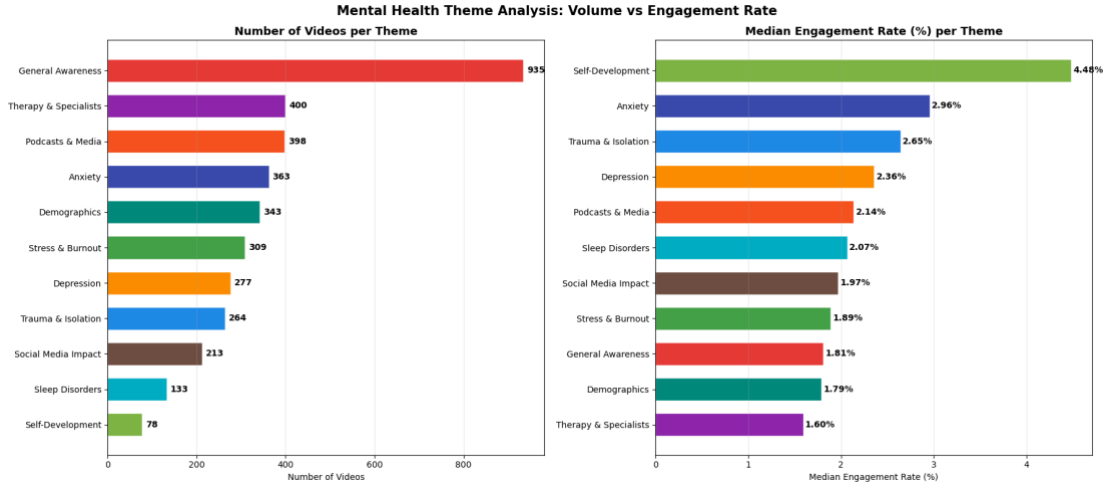
## 7.1.6 Correlation Analysis



A **Spearman rank correlation matrix** was computed across all numeric features to identify relationships between engagement metrics and influencer attributes. Spearman's method was chosen over Pearson's due to the non-normal distribution of the data and the presence of outliers. The heatmap revealed a strong positive correlation between views and likes ($r \approx 0.85$), indicating that content that reaches a wider audience consistently attracts more explicit approval interactions. A moderate correlation was also observed between views and comments, though weaker than the views-likes relationship, reflecting that passive engagement (likes) is more common than active engagement (comments) on both platforms.

### 7.1.7 Thematic Analysis

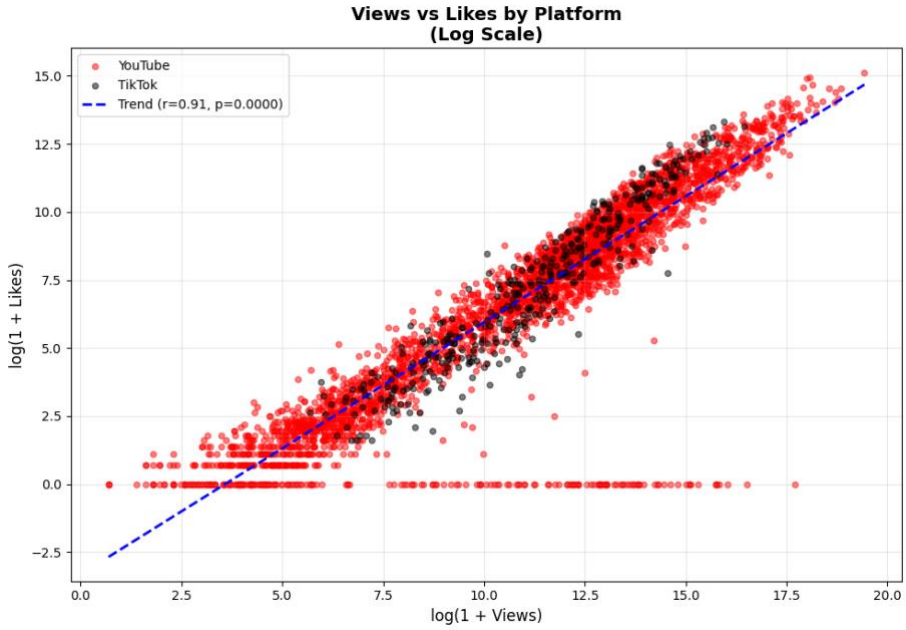
To identify the most prevalent mental health topics in the dataset, themes were assigned from the 57 unique search queries used during data collection, mapping each query to one of eleven thematic categories: General Awareness, Therapy & Specialists, Podcasts & Media, Anxiety, Demographics, Stress & Burnout, Depression, Trauma & Isolation, Social Media Impact, Sleep Disorders, and Self-Development.




General Awareness was the most produced category at 935 videos, followed by Therapy & Specialists (400 videos) and Podcasts & Media (398 videos). Condition-specific themes — Anxiety (363), Demographics (343), Stress & Burnout (309), Depression (277), and Trauma & Isolation (264) — each represents a smaller share of the content. Self-Development was the least produced theme at just 78 videos

The engagement metrics reveal a significant disparity between content production and audience reception. Self-Development content generated the highest median engagement rate at 4.48%, nearly double the dataset average of 2.75%. Anxiety content followed at 2.96%, and Trauma & Isolation at 2.64%. Conversely, Therapy & Specialists — despite being the second most produced category — recorded the lowest engagement rate at just 1.60%, and General Awareness, the most produced theme, came in near the bottom at 1.81%.

### 7.1.8 Engagement Rate

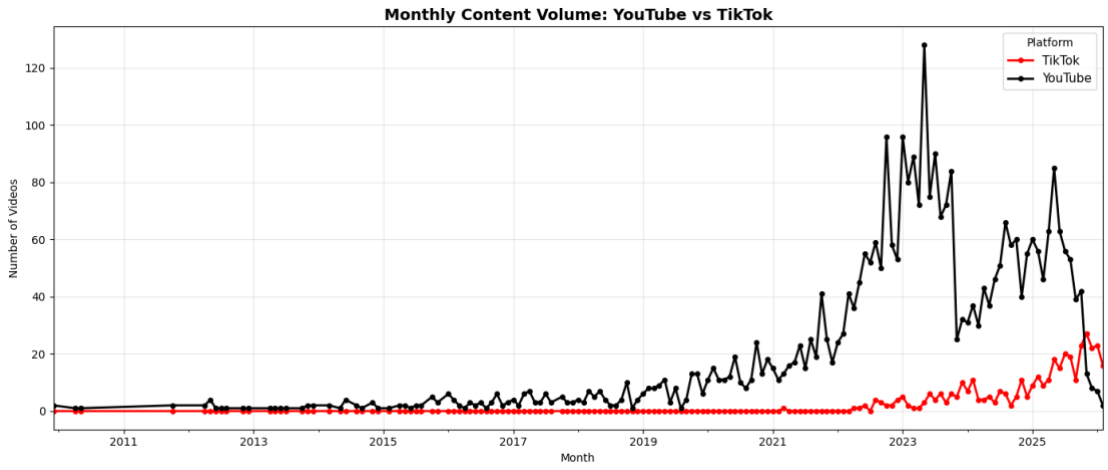



derived feature was engineered to enable normalized cross-platform comparison:

- **Engagement Rate (ER)** = (Likes + Comments) / Views × 100

A scatter plot of views versus likes, plotted on a log scale and colored by platform, revealed a clear linear trend in log-space, confirming the strong correlation identified in the Spearman analysis. A regression line was overlaid on the scatter plot, yielding a statistically significant fit ($p < 0.05$), with TikTok videos visibly clustering at higher engagement rate values relative to YouTube videos of comparable view counts.

### 7.1.9 Temporal Trends




To examine how content production evolved over time, video counts were aggregated by month and plotted separately for each platform. The temporal analysis revealed a

notable **increase in mental health content volume** across both platforms in recent periods, with YouTube maintaining a consistently higher volume of uploads. TikTok showed sharper month-to-month fluctuations, likely reflecting the platform's sensitivity to trending topics and viral content cycles. Seasonal spikes were observed during periods that may correspond to mental health awareness campaigns or significant social events in Saudi Arabia.

### 7.1.10 Summary of EDA Findings for primary data

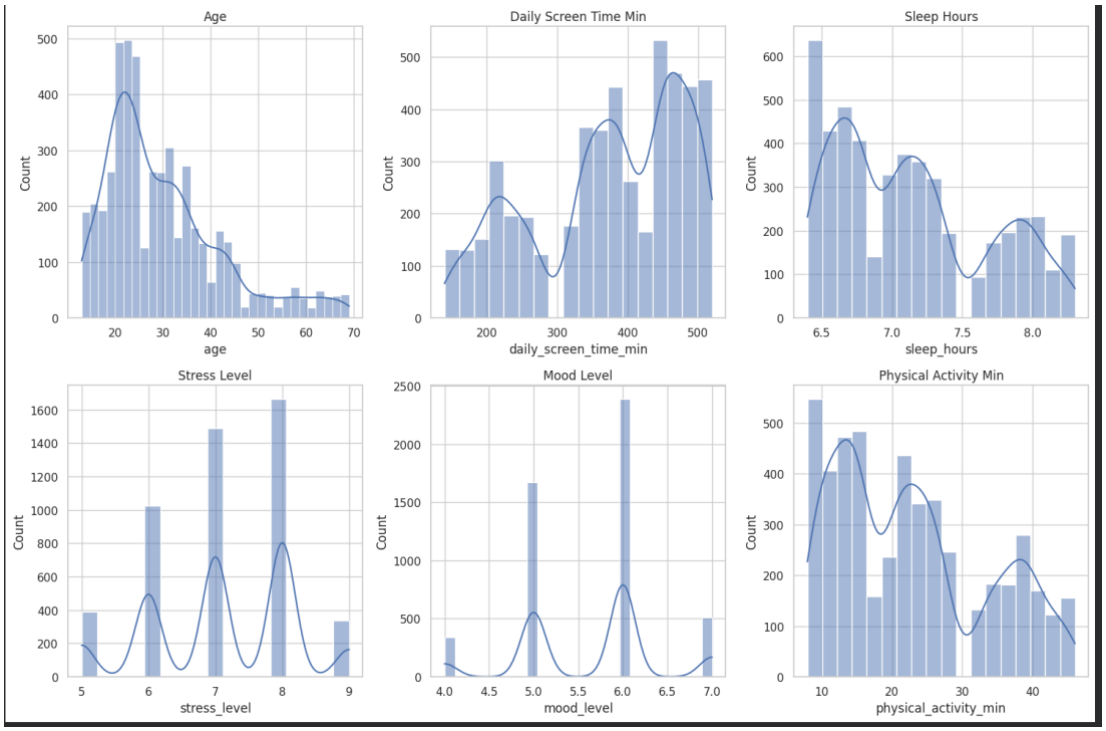
The exploratory analysis surfaced several meaningful patterns in the data. Engagement metrics are heavily skewed, with viral outliers requiring log transformation for meaningful analysis. TikTok achieves higher engagement efficiency despite lower content volume, while YouTube dominates in absolute content reach. Macro influencers produce the most content and attract the highest raw engagement, though Micro influencers show strong engagement rates relative to their size. The most produced mental health theme is general awareness, but the highest median engagement rate theme is self-development, and audience interaction is primarily passive (likes) rather than active (comments). These findings establish a solid foundation for the correlation and predictive modeling phases that follow.

## 7.2 Secondary Data

### 7.2.1 Overview of the Dataset

The dataset connects social media usage with mental health indicators. It covers the basics (age), usage stats (daily screen time, time on social media, interactions), and health factors (sleep, exercise). It also includes self-reported mental health scores for anxiety, stress, and mood. Plus, there are two custom metrics: screen_time_ratio (the share of screen time spent on social media) and wellness_score (a combined health rating).

### 7.2.2 Univariate Analysis



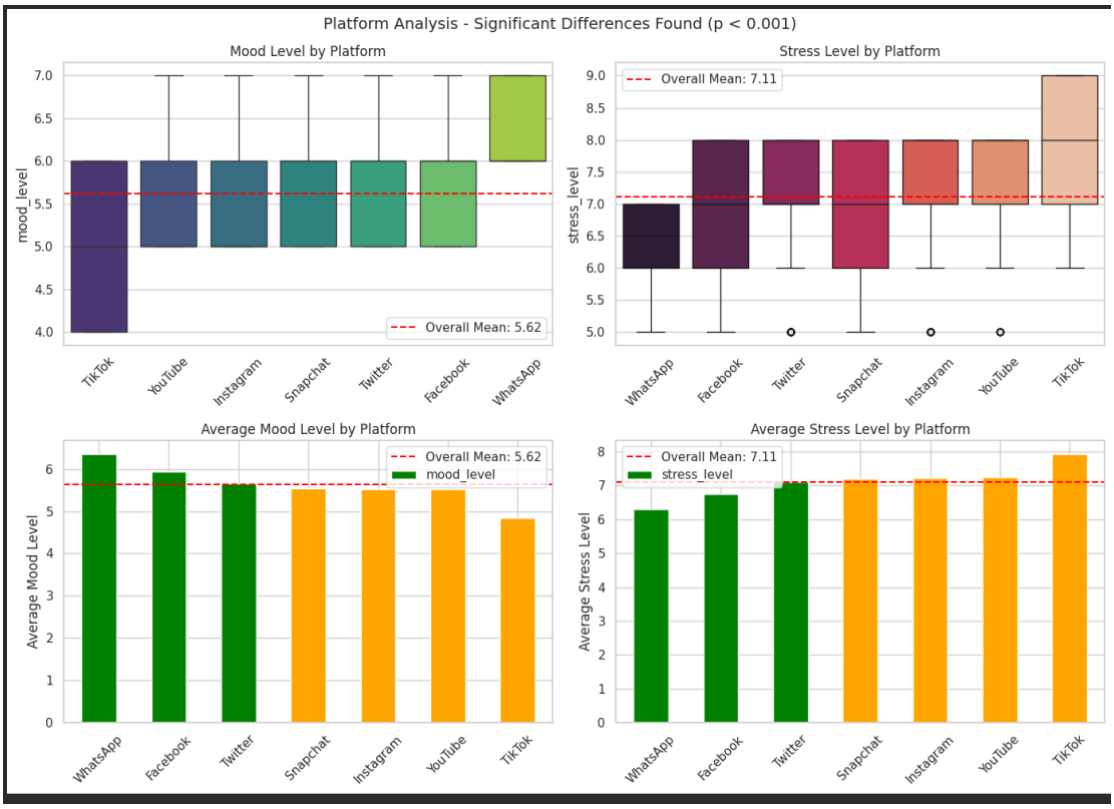

The univariate analysis gives a general idea about the distribution of the variables in the dataset. The age distribution shows that participants range from 13 to almost 70 years old, with most users in their 20s and 30s. Daily screen time varies widely between 140 and 520 minutes per day, which suggests that many users spend several hours a day on their devices.

Sleep hours mostly fall between about 6.5 and a little over 8 hours, which indicates that most people in the dataset gets enough hours of sleep. Stress levels appear relatively high overall, mostly between 6 and 8, while mood levels are generally around 5 to 6, suggesting a moderate mood level for most users.

Physical activity ranges roughly from 10 to 45 minutes per day, but many users seem to be closer to the lower end of this range. This suggests that a large portion of the sample has relatively low daily physical activity, which could potentially affect overall wellbeing.

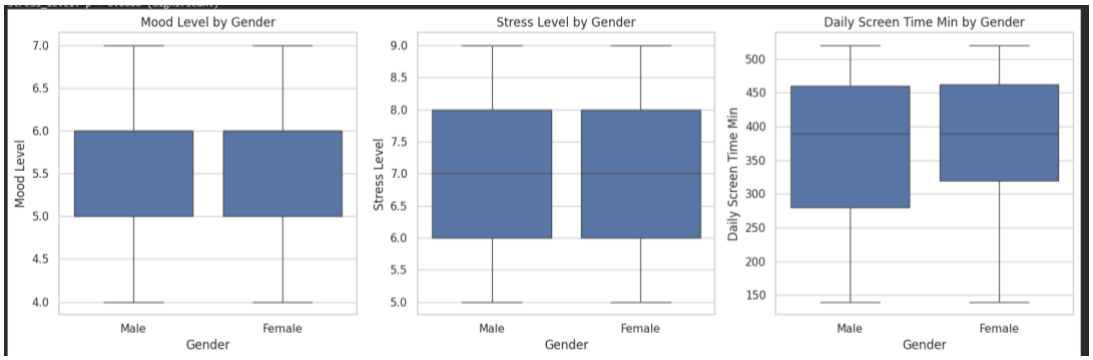
### 7.2.3 Group Comparisons

- Platform Comparison




The analysis shows noticeable differences in mood and stress levels across platforms. **WhatsApp users report the highest average mood (around 6.3)** and the **lowest stress levels (around 6.3)**, suggesting a more positive user experience. In contrast, **TikTok users report the lowest average mood (around 4.8)** and **the highest stress levels (around 7.9)**. Other platforms such as Facebook, Twitter, Instagram, Snapchat, and YouTube fall in the middle range with mood levels between 5.5 and 5.9 and stress levels around 6.7–7.1. These results suggest that platform usage may be associated with different mental health patterns, with WhatsApp linked to better mood and lower stress, while TikTok is associated with lower mood and higher stress.

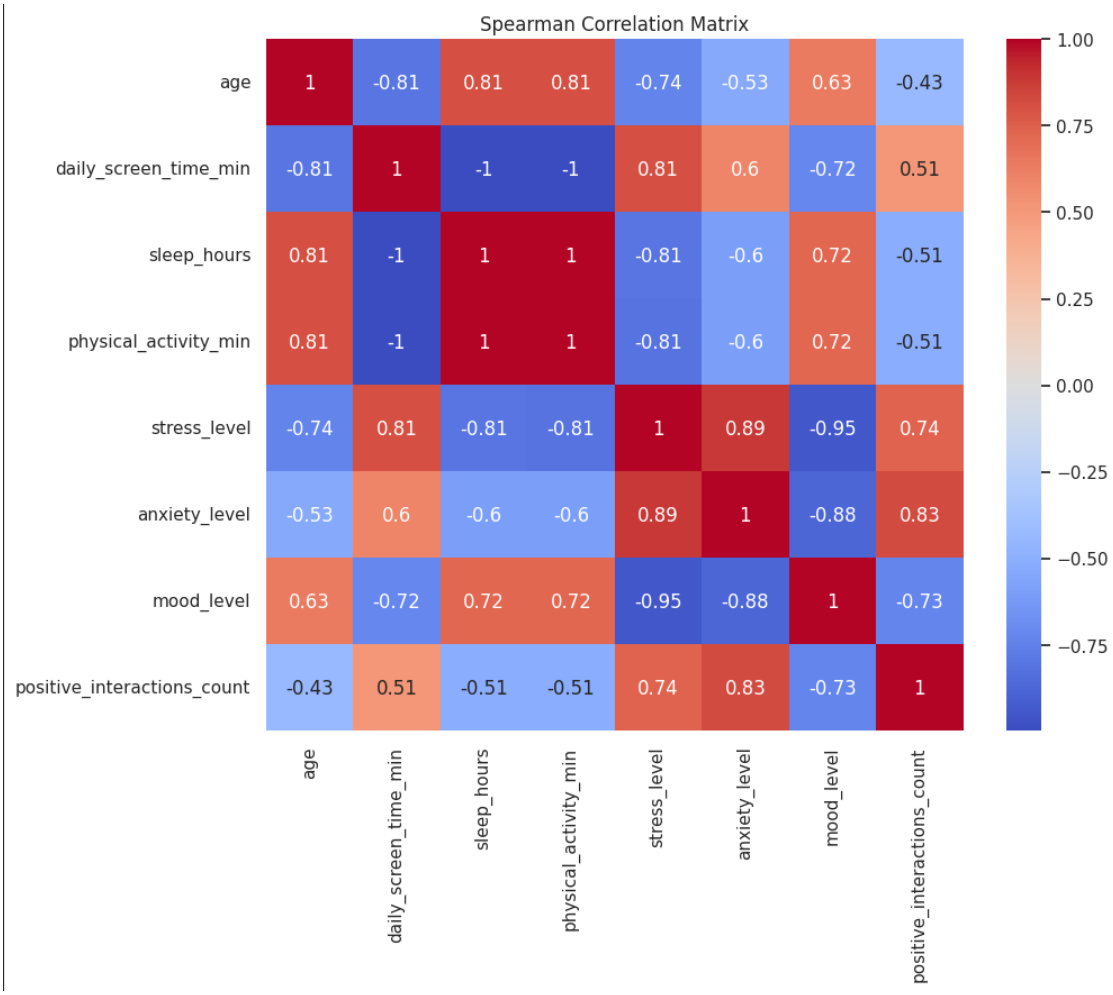
- Gender Comparison




A gender-based comparison was conducted using the Mann–Whitney U test to see if there are differences in mood level, stress level, and daily screen time. The results show that **there are no significant differences between males and females**. The p-values for mood level (0.4552), stress level (0.6031), and daily screen time (0.6976) are all higher than 0.05.

This means that the values for males and females are mostly similar. The boxplots also show this because the medians are close to each other and the ranges overlap between the two groups. So, **gender does not seem to have a strong effect on mood, stress, or daily screen time** in this dataset. On the other hand, other factors, like the social media platform people use, may have a bigger effect on mental health results.

## 7.2.4 Correlation Analysis




The Spearman correlation analysis highlights several strong relationships between lifestyle factors, social media usage, and mental health indicators. **Mood level shows a strong negative correlation with stress (-0.95) and anxiety (-0.88),** indicating that higher levels of stress and anxiety are associated with lower mood. In contrast, sleep hours and physical activity are positively correlated with mood (0.72), suggesting that healthier daily habits may support better emotional well-being.

**Daily screen time is negatively correlated with mood (-0.72) and positively correlated with stress (0.81)**, which may indicate that excessive screen use is linked to poorer mental health outcomes. Age also appears to influence lifestyle patterns, as it is positively correlated with sleep and physical activity (0.81) and negatively correlated with screen time (-0.81). Additionally, **positive interaction count** shows **positive correlations with stress (0.74) and anxiety (0.83) and a negative correlation with mood (-0.73)**, suggesting that individuals experiencing higher stress or anxiety may engage more frequently in online interactions.

### 7.2.5 Anomaly Analysis

The anomaly analysis reveals several notable patterns in the dataset. **No users were classified as having extreme screen time,** with the maximum daily screen time was found to be 520 minutes (8.7 hours). However, **60 users were found to have a 'High Screen Time + Low Mood' anomaly pattern**. Which indicates that a possible relationship exists between device usage and user moods. **57 users were flaged as 'At Risk'** based on various factors, including high stress levels, high anxiety, low moods, low sleep, and low physical activity levels. Such users could be a subject of further investigation since they form a small but important portion of 2.8% of the entire user base. The fact that several anomaly patterns are present in the data indicates that even though most users exhibit normal behavioral trends, a small but important portion of the user base shows worrying signs in their digital behavior and moods.

### 7.2.6 Summary of EDA Findings for secondary data

The exploratory analysis suggests that **lifestyle habits and social media usage may be related to mental health indicators**. For example, higher daily screen time seems to be associated with higher stress levels and lower mood. It was also observed that some social media platforms show higher stress and lower mood compared to others, which indicate that the type of platform a user spends time on can influence their mental health experience.

Based on these findings, a few assumptions can be suggested. **Platforms that rely more on short-form or fast content may be linked to higher stress and lower mood.** Another possible assumption is that exposure to supportive or educational content, such as posts shared by mental health influencers, may increase positive interactions and potentially help reduce stress or anxiety. These insights help provide a starting point for understanding how mental health influencers might affect public engagement across platforms.

However, this study has some **limitations** because it relies on secondary data. The dataset was not originally collected to study mental health influencers in Saudi Arabia, so it does not directly measure influencer content or real engagement with their posts. In addition, **some variables are self-reported and simplified**, which may limit how accurately the relationships can be interpreted.

# 8. Comparing Results of Datasets

## 8.1 Compare Key Metrics

(a) Compare Summary Statistics (Means, Medians, and Trends)

The primary dataset analyzes engagement metrics for YouTube and TikTok content.

The secondary dataset analyzes user-level indicators such as mood, stress, and screen time.

In the **primary dataset**:

- There were statistically significant differences between YouTube and TikTok in likes and views ($p < 0.05$).
- TikTok showed higher engagement efficiency despite having fewer videos.
- A strong positive correlation was found between views and likes ($r \approx 0.85$).

In the **secondary dataset:**

- There were statistically significant differences between platforms in mood and stress levels ($p < 0.001$).
- Average daily screen time was approximately 400 minutes.
- Stress had a strong negative relationship with mood ($r = -0.95$).
- Screen time was negatively related to mood ($r = -0.72$).

Both datasets show that platform differences are statistically significant, although they measure different outcomes.

(b) Create Tables Showing Metrics Side by Side

| Aspect | Primary Data | Secondary Data |
|---|---|---|
| Platform Comparison | Significant differences in likes and views ($p < 0.05$) | Significant differences in mood and stress ($p < 0.001$) |
| TikTok | Higher engagement efficiency | Higher reported stress levels |
| Strongest Correlation | Views–Likes ($r \approx 0.85$) | Stress–Mood ($r = -0.95$) |
| Screen Time | Not measured | Negatively associated with mood (-0.72) |
| Data Level | Video-level | User-level |

The table shows that while the primary dataset measures content engagement, the secondary dataset measures user well-being.

## 8.2 Contextualize Findings

(a) Use Secondary Data to Provide Broader Context or Validate Primary Data

The primary dataset showed that TikTok generates strong engagement compared to YouTube.

The secondary dataset showed that platform use is linked to significant differences in stress and mood levels.

This indicates that platform choice influences both interaction patterns and psychological outcomes, and shows that platform differences appear in both datasets, even though each dataset measures different outcomes.

(b) Explore Potential Reasons for Contradictions

The differences between the datasets can be explained by:

- The primary dataset measures engagement with content.
- The secondary dataset measures user mental health indicators.
- The primary data is based on videos, while the secondary data is based on individuals.

Because the two datasets measure different dimensions, differences in findings are expected.

# 9 Modelling

## 9.1 Baseline Model

To establish a performance benchmark prior to training more sophisticated models, a DummyRegressor was employed as the baseline. This model predicts the mean value of the target variable (engagement_rate) for all observations, regardless of the input features, making it a simple yet essential reference point for evaluating model improvement.

The dataset used for modelling was saudi_health_cleaned.csv, which contains the preprocessed and unified TikTok and YouTube records. The features selected for prediction were: year, month, duration_minutes, platform_encoded, duration_category_encoded, language_encoded, topic_encoded, sentiment_encoded, and influencer_tier_encoded. Categorical variables were encoded using LabelEncoder prior to modelling. Rows with missing values in any of these features were dropped, and the data was split into training (80%) and test (20%) sets using a fixed random seed (random_state=42) to ensure reproducibility.

The baseline model yielded an $R^2$ Score of 0.00 and an RMSE reflecting the natural variance of the target distribution confirming that predicting the mean engagement rate provides no explanatory power. This result serves as the lower-bound benchmark against which subsequent models Linear Regression and Random Forest Regressor are evaluated.

## 9.2 Model 2: Linear Regression

Following the establishment of the baseline, a Linear Regression model was implemented to explore the potential for a linear relationship between the predictors and the engagement rate. This model served as a primary statistical test to determine whether a constant rate of change between features could sufficiently explain the variance in engagement.

During the initial setup, the model faced practical difficulties because it cannot process missing values (**NaNs**). While the earlier baseline approach simply removed incomplete rows, it was observed that this led to a significant loss of useful data. To fix this, an important choice was made to use **median filling** across the features. This ensured the dataset remained complete, satisfying the model's requirements while keeping the data volume intact.

Upon evaluation, the Linear Regression model yielded an R^2 score of 0.02 and an RMSE of 3.07. This very small improvement over the baseline confirms that the factors driving social media engagement do not follow a simple linear path. The model’s failure to capture more than 2% of the variance suggests that engagement is influenced by complex interactions that a basic linear equation cannot represent.

## 9.3 Model 3: Random Forest Regressor

To address the limitations seen in the linear approach, the Random Forest Regressor was introduced as the third modeling phase. This algorithm was selected for its strength in handling large sets of data and its ability to capture complex, non-linear patterns by combining the results of multiple decision trees.

The model was set up with 200 trees to ensure a stable structure, using parallel processing to speed up the work. A major turning point in this phase was the adjustment of how categories were handled. It was found that the previous method (LabelEncoder) created a false numerical order between categories like platforms and topics, which confused the calculations. As a result, the team switched to One-Hot Encoding (using pd.get_dummies). This allowed the model to treat each category as an independent group, greatly improving its ability to tell the difference between various content attributes.

This adjustment led to a massive breakthrough in performance, with the Random Forest Regressor achieving an $R^2$ Score of 0.74 and a much lower RMSE of 1.59. This high predictive power shows that the forest-based approach, combined with the correct way of handling categories, successfully captured 74% of the factors that drive engagement rates.

## 9.4 Models Comparison and Selection

| Model | R2 Score | RMSE | MAE |
|---|---|---|---|
| Random Forest | 0.737540 | 1.588536 | 0.706538 |
| Linear Regression | 0.021027 | 3.067966 | 1. 802359 |
| Baseline (Dummy) | -0.000877 | 3.102098 | 1.863660 |

Three regression models were evaluated: the baseline model using DummyRegressor, Linear Regression, and Random Forest Regressor. The baseline model achieved an $R^2$ score of approximately -0.001 and an RMSE of 3.10, indicating that it simply predicts the mean and does not capture any meaningful patterns in the data. This serves as a reference point for evaluating more advanced models.

The Linear Regression model showed only a very slight improvement over the baseline, achieving an $R^2$ score of 0.02 and reducing the RMSE to 3.07. This indicates that there is almost no meaningful linear relationship between the input features (year, month, duration in minutes, platform, duration category, language, topic, sentiment, and influencer tier) and the target variable (engagement rate), and that the model struggles to learn useful patterns from the data.

The Random Forest model demonstrated the best performance among all models, with an $R^2$ score of 0.74 and the lowest RMSE of 1.59. This significant improvement suggests that the relationship between the features and the engagement rate is not purely linear, and that Random Forest, as a non-linear ensemble model, is better able to capture complex patterns and interactions within the data.

Based on these results, the Random Forest model was selected for production. It provides the highest explanatory power ($R^2$) and the lowest prediction error (RMSE), making it the most accurate and reliable model for regression prediction.

# 10.References

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